

PolyTrack UI

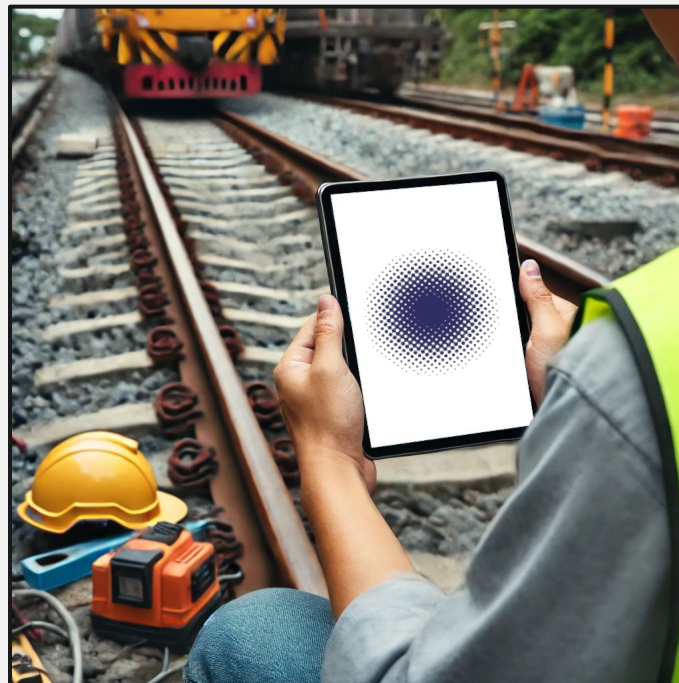
MVP

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Context & Vision

- **PolyChord Ltd** is a Cambridge University spin-out specializing in advanced data science solutions for rail maintenance.
- **PolyTrack** detects and ranks track defects using advanced sensors and data analysis.
- Current manual inspection methods are inefficient, unsafe, and reduce track lifespan.
- **Our focus:** developing a **user-friendly web app** for train maintenance crews to access fault data via mobile devices or desktops.



More about PolyChord at: <https://www.polychord.io>

User Stories



Maintenance Crew Members

Need a ***user-friendly app*** to locate and ***prioritize track faults*** for faster repairs.



Tram & Light Rail Companies

Want ***real-time updates*** on repair progress to ensure ***critical faults are addressed first***.

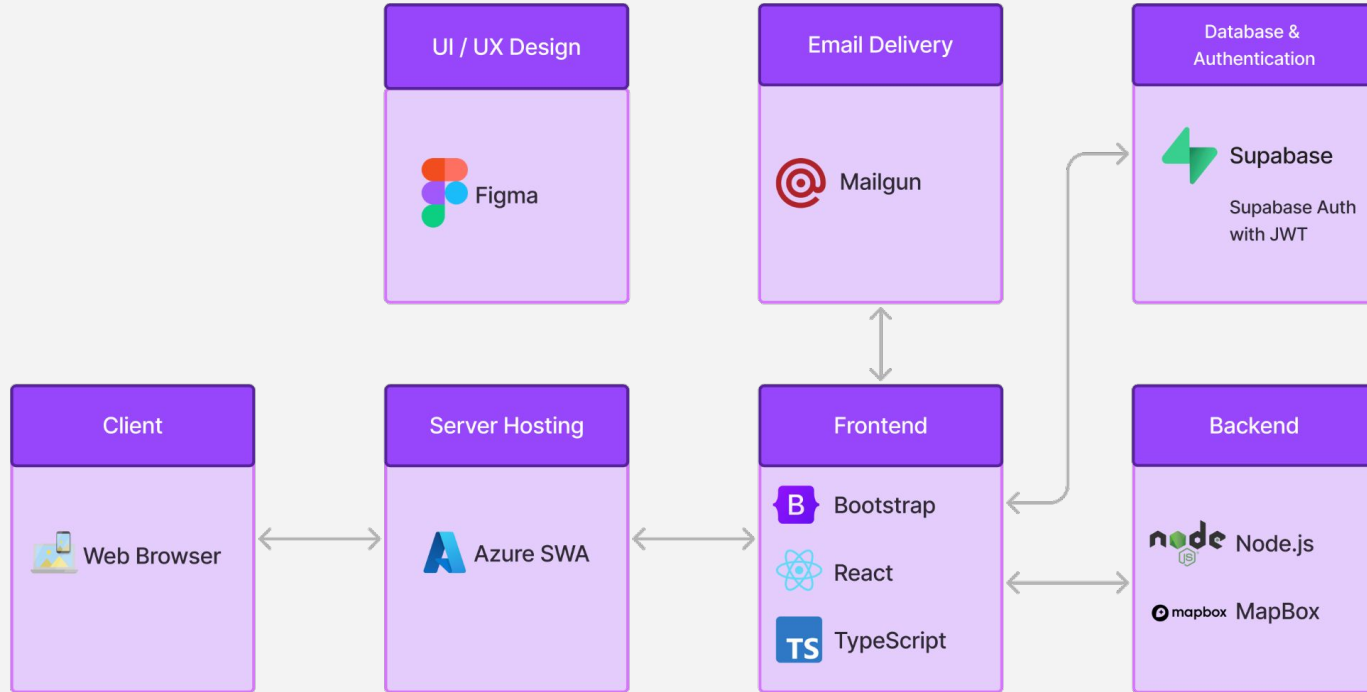


Operations Managers

Require ***immediate fault detection*** to minimize train service disruptions and improve ***passenger experience***.



Our Approach





Demo

<https://polytrack.tech>



Note: the app is currently accessible via any
@bristol.ac.uk email address

Achievements



10

Technologies
leveraged

CI/CD

Fully
operational

92.5%

Uptime during
testing

90+

GitHub issues
resolved

33 days

On the
leaderboard

26 days

With the Golden
Mouse

Client Feedback



UI/UX

They like the design of the pages and are impressed with how easy it is to use

Point of Interest Functionality

POI functionality is limited, they would like a way to see details about the issue.

Map View

The POIs on the map should be sorted based on rank in order to determine severity

Point of Interest Scope

Engineers should only see the issues in their area.



Challenges & Solutions



Problem

There isn't a nice or direct way of converting our Figma Pages into React.

Solution

Use the pages as a design guide and inspiration for our web pages.

Problem

There is a big resource and cost overhead to self-host Supabase directly on Azure.

Solution

Use the hosted version of Supabase included in the free tier of their software.

Problem

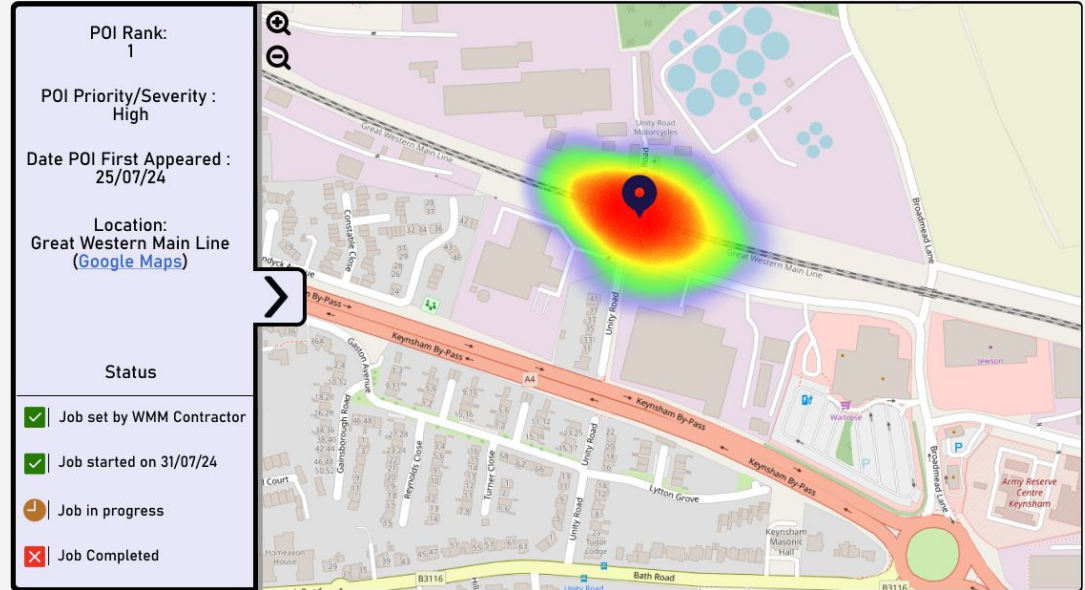
Some members of the team were working on less tasks than the others, causing an uneven workload.

Solution

Tasks are now assigned roughly equally to make sure everyone is working on at least one issue consistently.

Future Plans

- Refine the design of the website
- Colour POIs based on their rank
- Make POIs clickable to view the important details.
- Add a heatmap in the overview of the issue
- Allow the user to add in notes about an issue
- Allow the user to add an issue to the map





Thank you!

Any Questions?